

CATHERINE CHEN

she/her/hers ◇ US Citizen ◇ Updated Jan 2026
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RESEARCH INTERESTS

interpretability/explainability, information retrieval, natural language processing, large language models, human-centered AI, domain-specific applications

EDUCATION

Brown University

Ph.D Candidate in Computer Science

Sc.M in Computer Science

Advised by Carsten Eickhoff (now University of Tübingen)

Providence, RI, USA

Sept 2021 - Present

May 2023

University of Amsterdam

Visiting Researcher

Hosted by Maarten de Rijke and Maria Heuss (IR Lab)

Amsterdam, Netherlands

Feb - May 2025

Wellesley College

B.A. in Computer Science and French

Wellesley, MA, USA

Aug 2015 - May 2019

PUBLICATIONS

**Denotes equal contribution*

- Yumeng Wang, **Catherine Chen**, and Suzan Verberne. *RankSteer: Activation Steering for Pointwise LLM Ranking*. Under review 2026.
- Yumeng Wang, Jirui Qi, **Catherine Chen**, Panagiotis Eustratiadis, and Suzan Verberne. *How Role-Play Shapes Relevance Judgment in Zero-Shot LLM Rankers*. Accepted to ECIR 2026.
- Meng Lu*, **Catherine Chen***, and Carsten Eickhoff. *Pathway to Relevance: How Cross-Encoders Implement a Semantic Variant of BM25*. EMNLP 2025.
- Gregory Polyakov, **Catherine Chen**, and Carsten Eickhoff. *Towards Best Practices in Axiomatic Activation Patching in Information Retrieval*. SIGIR 2025.
- Andrew Parry, **Catherine Chen**, Carsten Eickhoff, and Sean MacAvaney. *MechIR: A Mechanistic Interpretability Framework for Information Retrieval*. ECIR 2025.
- **Catherine Chen**, Jack Merullo, and Carsten Eickhoff. *Axiomatic Causal Interventions for Reverse Engineering Relevance Computation in Neural Retrieval Models*. SIGIR 2024.
- **Catherine Chen** and Carsten Eickhoff. *Evaluating Search System Explainability with Psychometrics and Crowdsourcing*. SIGIR 2024.
- William Rudman, **Catherine Chen**, and Carsten Eickhoff. *Outlier Dimensions Encode Task Specific Knowledge*. EMNLP 2023.
- **Catherine Chen**. *Quantifying and Advancing Information Retrieval System Explainability*. SIGIR 2023.

WORK EXPERIENCE

Centre for the Governance of AI

Summer Fellow

London, UK

June - Aug 2025

- Designed a framework to operationalize explainability auditing, mapping regulatory transparency goals to granular explanation properties and existing XAI tools.

Blue Cross and Blue Shield of Minnesota

Data Science PhD Intern, AI Models and Platform Team

Remote

June - Sept 2024

- Collaborated with healthcare professionals to develop an end-to-end pipeline leveraging LLMs using AWS services (Sagemaker, Bedrock, S3) for automated Behavioral Health patient case summarization.

FreeWheel, A Comcast Company*Software Engineer, Creative Team*

New York, NY, USA

Aug 2019 - Aug 2021

- Implemented backend APIs in Golang with gRPC to ingest and manage advertising creatives stored in AWS S3/MySQL DB, supporting large-scale ad inventory workflows.
- Constructed user interface components for a React-based web application within a service-oriented, containerized backend architecture using Docker and Kubernetes.

PatientsLikeMe*Software Engineering Intern*

Cambridge, MA, USA

June - Aug 2018

- Developed features for a large-scale web application using Ruby on Rails and React, supporting a personalized health platform for individuals with chronic conditions.

University of Massachusetts, Amherst*Research Assistant, Knowledge Discovery Lab*

Amherst, MA, USA

June - Aug 2017

- Reverse-engineered Google Deepmind's DQN to identify influential neurons to explain how the network learned to play Atari Breakout.

MENTORSHIP

Master's Students

Meng Lu - Brown University

July 2024 - Sept 2025

Kora Helm - University of Tübingen

*Sept 2023 - July 2024***Undergraduate Students**

Pearl Zhang – Swarthmore College – with Prof. Jeff Huang

Jan - May 2023

Yuhan Wang – Smith College → Georgia Tech PhD – with Prof. Jeff Huang

Jan - May 2023

Lucas Weissman – Williams College – with Roma Patel

June - Aug 2022

Marwa Bouabid – Mount Holyoke – with Prof. James Tompkin

*Jan - May 2022***TEACHING**

Brown UniversityCSCI 1460: Computational Linguistics, *Graduate TA**Sept - Dec 2024*Sheridan Center for Teaching and Learning, *Teaching Certificate I**Sept - Dec 2023*NSF REU: AI for Computational Creativity, *Research Methods Study Group Leader**June - Aug 2022, 2023***Wellesley College**CS 235: Theory of Computation, *Supplemental Instructor**Sept - Dec 2018***INVITED TALKS**

Peeking Under the Hood: Interpretability and Explainability in IR*European Summer School on Information Retrieval (ESSIR)**July 2025***Reverse Engineering Mechanisms of Relevance in Neural IR Models***University of Amsterdam**May 2025**Radboud University**May 2025**Leiden University**April 2025***Advancing Explainable Information Retrieval: Methods for Human-Centered Evaluation and Interpreting Neural IR Models***University of Glasgow**Sept 2024***Choosing Between Industry and Academia***Brown University, ExploreCSR: Diversity in Computer Systems**April 2023***Women in STEM Workshop: Pursuing a Career in CS***The Stemistry Project**Feb 2023*

LEADERSHIP/SERVICE

Faculty-Grad Liason

Department of Computer Science

Brown University

Fall 2024 - Present

- Communicate with the department about graduate student affairs/concerns and coordinate student volunteer positions.

CS Department Representative

Graduate Student Council

Brown University

Sept 2022 - May 2024

- Attended monthly meetings to facilitate collective action on graduate student-related issues.

Graduate Student Orientation and Recruitment

Department of Computer Science

Brown University

Jan 2022 - Mar 2024

- Collaborated with department administrators to organize welcome events for admitted/incoming students.

Graduate Community Fellow

Graduate School

Brown University

Sept 2022 - May 2023

- Designed and implemented community-building initiatives for Women and Non-Binary Graduate Students to celebrate diversity, promote belonging and inclusion, and connect individuals across disciplines.

Co-Organizer

Tutorial on Mechanistic Interpretability @ ECIR '26

Mar 2026

Workshop on Explainability in Information Retrieval (WExIR) @ SIGIR '25

Jul 2025

New England NLP Meetup

Apr 2024

Conference Reviewer

Actionable Interpretability Workshop

2025

ACL Rolling Review (*EMNLP '24 Outstanding Reviewer*)

2024 - 2025

CHIIR

2026

ECIR

2024 - 2026

SIGIR

2024 - 2025

WSDM

2025

Panelist

Amplify: Students as Leaders: At Wellesley and Beyond

Oct 2024

Wellesley CS Club: Wellesley Careers in Tech Panel

Apr 2024

AlumConnect: Research at Wellesley

Feb 2024

AI/ML Panel at Hack @ Brown

Feb 2024

Moderator

Grad Student Discussion Panel for REU AI @ Brown

Aug 2023

Grad Student Discussion Panel for ExploreCSR AI @ Brown

Apr 2023

Women and Non-Binary Faculty Panel @ Brown

Mar 2023

Other Service

Max Planck Institute for Intelligent Systems PhD Application Reader

Nov 2024

Brown CS PhD Application Reader

Jan 2024

AWARDS

Research Mobility Fellowship, Brown University

Spring 2025

Graduate Community Fellow, Brown University

Fall 2022 - Spring 2023

SKILLS

Programming

Python, React, JavaScript, Golang, Ruby on Rails, MySQL, HTML, CSS

Practices/Tools

PyTorch, Git, Pandas, AWS Services (Bedrock, Sagemaker, S3), Docker, Kubernetes, Flask, Jira, Confluence, Agile, Scrum

Languages

English, French